QUESTION PAPER

Model Paper — 04

Mathematics II

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Time: 3 hours (plus 10 minutes reading time) **Total marks:** 100

Index number:

Important:

- Select **five** questions from Part A and **five** questions from Part B, and answer ten questions in all.
- Show the relevant steps and write the correct units in your answers.
- Each question carries 10 marks.
- Write your answers on separate answer sheets. Where a question says “copy the figure”, copy it onto your answer sheet.
- Mathematical tables may be used. In constructions use only a straight edge with a cm/mm scale and a pair of compasses.
- Volume of a right pyramid of base area A and height h = $\frac{1}{3}$Ah; volume of a right circular cylinder of base radius r and height h = $\pi r^2 h$.

Part A

Answer only five questions. (10 marks each)

1. Lahiru plans to buy a new oven for his bakery. The cash price of the oven is Rs 96 000. A finance company offers two methods of payment, each over a period of six months.

Method 1: Take a loan of Rs 96 000 and settle it, together with the total interest of Rs 10 080, in one payment at the end of the six months.

Method 2: Repay the Rs 96 000 in 6 equal monthly parts of capital, one at the end of each month. Interest is charged every month on the balance owed at the beginning of that month (reducing balance). The total interest paid over the six months is also Rs 10 080.

(i) Find the annual interest rate of Method 1. (2 marks)

(ii) (a) Write down the balance owed at the beginning of each of the six months, and find the sum of these six balances. (3 marks)

(b) Using this, find the annual interest rate of Method 2. (3 marks)

(iii) Show that Method 2 has the higher annual interest rate, and give a reason for this. (2 marks)

2. The table below shows some values of the function $y = f(x)$ for $-2 \leq x \leq 6$. The graph of the function is symmetric about the line $x = 2$.

x	−2	−1	0	1	2	3	4	5	6
y	−7	0	5	8	9		5	0	−7

(i) Using the symmetry of the graph, find the missing value of y in the table. (1 mark)

(ii) Using a suitable scale on both axes, draw the graph of the function on graph paper. (3 marks)

(iii) (a) Write down the x-coordinates of the points where the graph cuts the x-axis. (1 mark)

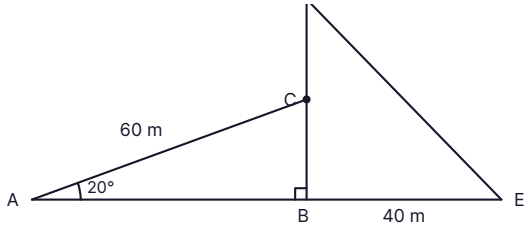
(b) Hence write the function in the form $y = -(x + p)(x + q)$, where p and q are constants. (1 mark)

(iv) The straight line $y = 3$ cuts the graph at the points A and B. Using the graph, (a) write the range of values of x for which $y > 3$; (2 marks)

(b) find the length of AB, to the nearest first decimal place. (2 marks)

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3. The figure shows the side view of a railway embankment. A, B and E are on level ground, with B between A and E, and BD is vertical. C is a point on BD such that DC = BC. The ramp AC is 60 m long, BE = 40 m and ∠CAB = 20°. **Copy the figure onto your answer sheet and mark the given data on it.**



Using trigonometric tables, show that the angle of inclination of DE, ∠DEB, is greater than ∠CAB by more than 25°.

(i) Find the length of BC. (3 marks)

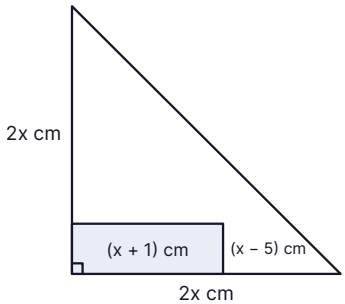
(ii) Find the length of BD, and hence the value of tan ∠DEB. (3 marks)

(iii) Find the size of ∠DEB, using tables. (2 marks)

(iv) Hence complete the proof. (2 marks)

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4. Rashmi decorates the top of a cake whose surface is a right-angled isosceles triangle with equal sides 2x cm. A rectangular space with sides (x + 1) cm and (x − 5) cm is marked inside it for a greeting, as shown in the figure. The remaining part is decorated with icing, and its area is 100 cm^2.



(i) Show that x satisfies the equation $x^2 + 4x - 95 = 0$. (4 marks)

(ii) Solve the equation, and find the value of x to the nearest whole number of centimetres. (Take $\sqrt{11} = 3.32$.) (3 marks)

(iii) Using the value of x found in (ii), show that the longest edge of the cake top, $2\sqrt{2}$ x cm, is greater than 22 cm. (Take $\sqrt{2} = 1.41$.) (3 marks)

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5. Dinuka, who lives in a fishing village, bought 120 baskets in all for storing fish, some small and some large, for Rs 12 450. A small basket costs Rs 85 and a large basket costs Rs 130.

(i) Taking the number of small baskets as x and the number of large baskets as y, form a pair of simultaneous equations, and solve them to find the number of baskets of each kind. (5 marks)

(ii) A large basket holds 4 kg more fish than a small basket. When all the baskets are full, they hold 920 kg of fish altogether. Find the number of kilograms of fish a small basket holds. (5 marks)

6. Chamari records the number of loaves of bread sold per day at her bakery over 40 days. The data are given in the table.

Loaves sold per day	40 – 49	50 – 59	60 – 69	70 – 79	80 – 89	90 – 99
Number of days	2	7	10	12	6	3

(i) Find the mean number of loaves sold per day. (3 marks)

(ii) The bakery is open for 26 days in a month. Find the expected total number of loaves sold in a month. (2 marks)

(iii) The profit from one loaf is Rs 14. Find the expected profit in a month. (3 marks)

(iv) Find the minimum possible total number of loaves sold on the days on which fewer than 60 loaves were sold. (2 marks)

Part B

Answer only five questions. (10 marks each)

7. Use only a straight edge with a centimetre/millimetre scale and a pair of compasses. Construction lines must be clearly visible.

(i) Construct the line segment AB = 5 cm. At B, construct ∠ABX = 150° (by constructing angles of 90° and 60°). (3 marks)

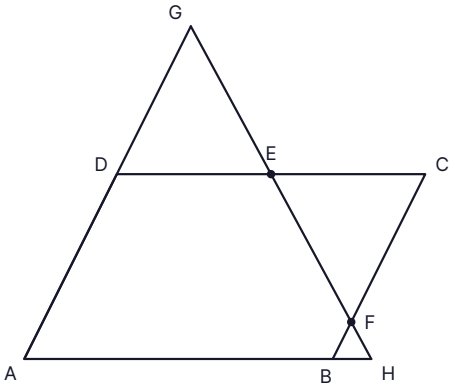
(ii) Mark the point C on BX such that BC = AB, and join AC to form the isosceles triangle ABC. (1 mark)

(iii) Construct the perpendicular bisector of AC, and with AC as the diameter construct a semicircle on the side of AC opposite to B. (3 marks)

(iv) Construct the line through A parallel to BC, meeting the semicircle at D. Join CD. (2 marks)

(v) Write down the size of ∠ACD. (1 mark)

8. In the parallelogram ABCD, E is the midpoint of CD, and F is a point on BC such that BF = $\frac{1}{5}$ BC. FE produced meets AD produced at G, and FE produced the other way meets AB produced at H. **Copy the figure onto your answer sheet.**



(i) Show that the triangles DGE and CFE are congruent. (3 marks)

(ii) Join DF and CG. Show that DGCF is a parallelogram. (2 marks)

(iii) Show that the triangles BFH and DGE are equiangular. (2 marks)

(iv) Show that BH = $\frac{1}{8}$ AB. (3 marks)

9. Lahiru melts 15 identical solid metal pyramids, each with a square base of side a and height 3a, without any loss, and casts a solid right circular cylinder of base radius r and height 5r from the metal.

(i) Show that $a^3 = \frac{\pi}{3} r^3$. (4 marks)

(ii) When r = 7.6 cm and π = 3.14, find the value of a^3 using logarithms, and hence find the value of a to the nearest centimetre. (3 marks)

(iii) A solid cuboid has a square base of side a and height 3a. A pyramid with the same base and height as the cuboid is cut out of it. Using a = 8 cm, find the volume of the metal remaining. (3 marks)

10. The first track-laying gang of a railway company lays 30 m of new track on the first day, and each day after that it lays 4 m more than on the previous day. It completes the whole job on the 12th day.

(i) Write down the lengths of track laid on the first three days. (1 mark)

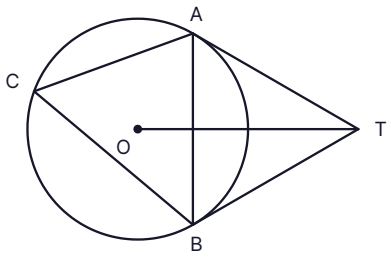
(ii) Find the length of track laid on the 12th day. (2 marks)

(iii) Find the total length of the track. (2 marks)

(iv) A second gang lays the same total length. It lays 6 m more each day than on the previous day, and it completes the job on the 13th day. Find the length it lays on the first day. (3 marks)

(v) On which day do the two gangs lay the same length of track? (2 marks)

11. In the figure, O is the centre of the circle. TA and TB are tangents to the circle at A and B respectively, and C is a point on the major arc AB. **Copy the figure onto your answer sheet and join OA and OB.**



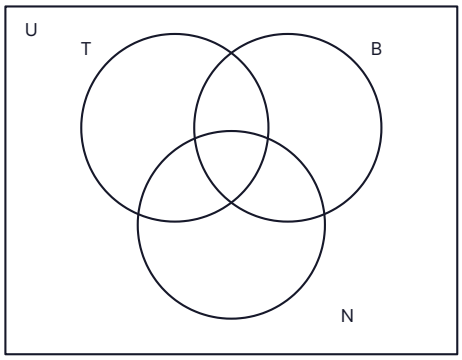
(i) (a) Show that OATB is a cyclic quadrilateral. (3 marks)

(b) Show that OT bisects ∠AOB. (2 marks)

(ii) Name two angles equal to ∠TAB, giving a reason for each, and hence show that ∠ATB + 2∠ACB = 180°. (5 marks)

12. A survey was done on 120 passengers at a railway station kiosk. The items they bought were tea (T), buns (B) and newspapers (N). The figure shows the universal set and the three sets. **Copy the figure onto your answer sheet and fill in the data.**

12 passengers bought none of the three items, and 36 passengers bought two or more kinds of items.



(i) Find the number of passengers who bought exactly one kind of item. (3 marks)

(ii) If the number of passengers who bought all three items is equal to the number who bought exactly two kinds of items, find the number who bought all three items. (3 marks)

(iii) The number who bought only tea and buns is 6. If the number who bought only buns and newspapers is twice the number who bought only tea and newspapers, find each of these two numbers. (2 marks)

(iv) If 56 passengers bought tea and 50 bought newspapers, find the number who bought only buns. (2 marks)

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